

Revision Diff Report: v1 $\rightarrow$ v2

Paper: *Volatility Targeting in the Taiwan Stock Market*

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Revision by: VolPred Research System (Claude Opus 4.6)

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1 Overview

Metric	v1	v2
Page count	45	43
Main-body contributions	3	2
Tables (main body)	7	8 (+1 new common-period table)
Figures (main body)	2	2
Appendix sections	0	1 (TZ analysis, 6 subsections)
Appendix tables	0	1 (TZ table moved)
Appendix figures	0	1 (Fig. 3 moved)

Three HIGH-severity issues from the v1 academic review were addressed:

1. **H1**: Non-comparable evaluation periods (Table 3)
2. **H2**: Scope overreach — three loosely connected contributions
3. **H3**: Diversification amplification based on $N = 10$ (including 1 ETF)

2 H1: Non-Comparable Evaluation Periods

Problem. Table 3 (v1) compared strategies across different time periods (BH 2010–2026, GJR VT 2020–2026, 8.63/VIX 2016–2026), making Sharpe ratio rankings unreliable.

Changes:

- **[+]** New Table 4 (“Common-Period Strategy Comparison, 2020–2026”) evaluates *all five* strategies over the identical 2020–2026 window. Over this common period:

- GJR VT: 1.108 (unchanged, this was already its native period)
- GARCH VT: 0.994
- EWMA VT: 0.927 (up from 0.796 over 2010–2026)
- 8.63/VIX: 0.856 (up from 0.690 over 2016–2026)
- Buy & Hold: 0.824 (up from 0.729 over 2010–2026)
- [\[M\] Table 3 notes](#) updated with cross-reference to Table 4 and stronger caveat language.
- [\[+\] Explanatory paragraph](#) between Tables 3 and 4 documents that the Sharpe gap between GJR VT and 8.63/VIX narrows from 0.418 to 0.252 over the common period, confirming that period effects explain a substantial portion of the apparent superiority.
- [\[M\] Abstract, Introduction, Conclusion](#) updated to reference common-period results.

3 H2: Scope Overreach — Three to Two Contributions

Problem. The time-zone (TZ) information transmission analysis (Section 5 in v1) read as a separate paper grafted onto the VT paper.

Changes:

- [\[-\] Section 5 \(Time-Zone Information Transmission\)](#) removed from main body.
- [\[+\] Appendix A](#) created with the full TZ analysis (6 subsections, 1 table, 1 figure), labeled as “supplementary evidence.” Content is preserved intact.
- [\[M\] Abstract](#) rewritten from “Three contributions” to “Two contributions.” TZ channel mentioned as supplementary evidence. Keywords updated (removed “cross-market price discovery,” “opening auction efficiency”; removed G15 from JEL).
- [\[M\] Introduction](#) restructured:
 - Contribution 1 (leverage amplification): unchanged in substance, updated numbers.
 - Contribution 2 (VT strategies): expanded to include cross-market validation (“2 of 4 U.S. findings generalize”). Common-period comparison referenced.
 - Former Contribution 3 (TZ): replaced with a brief paragraph pointing to Appendix A.

- [M] Section ordering changed: former Sections 6 (Macro) and 7 (VaR) are now Sections 5 and 6, flowing directly from VT strategies.
- [M] Discussion section:
 - [–] Subsection “ETFs Versus Individual Stocks for Momentum Strategies” removed (content moved to Appendix A.6).
 - [–] Practical implications: “Time-zone momentum” and “Combined strategy” items removed from main body (covered in Appendix).
 - [M] “Comparison to Prior Literature” shortened: TZ discussion replaced with brief pointer to Appendix.
- [M] Conclusion rewritten from three findings to two findings, with a brief paragraph pointing to Appendix for TZ evidence.
- [M] All cross-references from “Section 5” to “Appendix A” throughout.

Net effect: Main body reduced by ~ 2 pages. The TZ analysis is fully preserved in the appendix, available to referees who want the details.

4 H3: Diversification Amplification ($N = 10 \rightarrow N = 9$)

Problem. The $4.6\times$ amplification ratio was based on only 10 securities, one of which (0056.TW) is an ETF misclassified as an “individual stock.”

Changes:

- [M] 0056.TW reclassified. Data section now reads “Nine individual Taiwan stocks and one ETF.” 0056.TW labeled as ETF throughout.
- [M] Tables 1 and 2 updated:
 - 0056.TW separated into its own row under “ETF (excluded from individual-stock average).”
 - New row: “9-stock average (excl. 0056)” with $\bar{\gamma} = 0.054$.
 - Existing row relabeled: “10-security average (incl. 0056 ETF)” with $\bar{\gamma} = 0.060$.
 - Table notes explain why 0056 is excluded from the primary average.
- [M] Amplification ratio recalculated:
 - Primary: $0.272/0.054 = 5.0\times$ (9 stocks, excluding 0056 ETF)
 - Secondary: $0.272/0.060 = 4.5\times$ (10 securities, including 0056 ETF)
 - 0050.TW-based: $0.087/0.054 = 1.6\times$ (was $1.45\times$ with 0056)

- [\[+\]](#) [Bootstrap confidence interval](#). New paragraph reports 90% CI from block bootstrap (10,000 reps, block = 252 days): [2.8, 8.1] with median 4.9. The CI excludes 1.0 but includes the U.S. benchmark at its lower bound.
- [\[+\]](#) [“Sensitivity to 0056.TW inclusion” paragraph](#). Documents that including 0056 biases the ratio downward (conservative direction) and is factually incorrect.
- [\[M\]](#) [Language downgraded](#): “4.6 $\times$ ” $\rightarrow$ “approximately 5.0 $\times$ (suggestive evidence)” throughout abstract, introduction, discussion, and conclusion.
- [\[M\]](#) [Abstract](#) now includes CI and N in parenthetical.
- [\[M\]](#) [Conclusion](#) explicitly acknowledges wide confidence interval as a limitation.

5 Other Minor Changes

- [\[-\]](#) [Two Codex inline comments](#) (“Codex-R6-fix”) removed from body.
- [\[M\]](#) [Fixed missing space](#) before “provides statistical evidence” (line 223).
- [\[+\]](#) [Appendix TZ table](#) wrapped in `resizebox` to fix overfull hbox.

6 Summary of Changes by Location

Location	Type	Description
Abstract	[M] Rewrite	3 $\rightarrow$ 2 contributions; added CI and N ; removed TZ; updated keywords/JEL
Introduction §1	[M] Edit	Updated amplification numbers; restructured contributions; TZ $\rightarrow$ appendix pointer
Data §2.1	[M] Edit	“Ten individual stocks” $\rightarrow$ “Nine individual stocks and one ETF”
Table 1	[M] Edit	Added 9-stock average row; relabeled 10-security row; updated notes
Table 2	[M] Edit	0056 separated; 9-stock average added; notes updated
§3.2 (Amplification)	[M] Rewrite	New CI paragraph; sensitivity paragraph; language downgrade

Location	Type	Description
Table 3 notes	[M] Edit	Cross-reference to new Table 4
Table 4	[+] New	Common-period comparison (2020–2026)
§5 (TZ)	[-] Moved	Entire section moved to Appendix A
§6 → §5	[M] Renumber	Macro indicators
§7 → §6	[M] Renumber	VaR and risk management
Discussion	[M] Edit	Removed TZ-specific subsections; shortened lit comparison
Conclusion	[M] Rewrite	3 → 2 contributions; added CI limitation; TZ as appendix pointer
Appendix A	[+] New	TZ analysis (6 subsections, 1 table, 1 figure)

7 Files

- `body.tex` — v1 body (unchanged, preserved as reference)
- `body_v2.tex` — v2 body with all three fixes
- `main.tex` — v1 driver (inputs `body.tex`)
- `main_v2.tex` — v2 driver (inputs `body_v2.tex`; updated abstract)
- `main_v2.pdf` — compiled v2 (43 pages)
- `v1_to_v2_diff.tex` — this document